

Kumar Gaurav

Software Engineer | AI/ML Systems & Backend

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linkedin.com/in/kumar-gaurav-cs | github.com/D3S-Gaurav | Codeforces: ATHEUS | LeetCode: ATHEUS_007

Skills

Languages: Python, TypeScript, JavaScript, Go, SQL, Bash, C++ (competitive programming)

ML & Data: scikit-learn, LightGBM, pandas, NumPy, SciPy, feature engineering, cross-validation, NLP (TF-IDF)

LLM Systems: multi-agent orchestration, structured outputs (Zod), prompt-injection defense, Gemini API, Temporal

Backend: FastAPI, Node.js, Express.js, Next.js, React, Socket.IO, WebSockets, gRPC, Kafka, REST APIs

Data & DevOps: PostgreSQL, Redis, Prisma, Docker, GitHub Actions (CI/CD), Git, Linux

Testing: k6 (load testing), Vitest, Supertest, Go testing (race detector)

Education

Birla Institute of Technology, Mesra | Ranchi, Jharkhand

Aug 2023 – May 2027 (Expected)

B.Tech in Computer Science and Engineering, CGPA: 7.88/10

Experience

Backend Engineer | SoilSense (Soil Suitability Platform)

Mar 2026 – Present

Team project, Schneider Electric Hackathon 2026 | Python, FastAPI, pandas, Next.js, Gemini API | GitHub

- Wrote the crop-suitability rules engine over a 2,200-sample, 22-crop dataset, with tolerance bands (10% on NPK, 0.3 on pH) that map each deficiency to an organic amendment.
- Built an all-crops endpoint that loads the dataset once instead of once per crop, making the 22-crop scan **2.3× faster** (48 ms to 21 ms) and replacing 22 client requests with one.
- Hardened the Gemini analysis route against prompt injection by allow-listing and type-casting the 5 expected sensor fields, so injected keys never reach the prompt.

Open Source Contributor | Oppia, FOSSology

Jan 2026 – Present

3 merged upstream PRs | TypeScript, Angular, ESLint, CI/DevOps | Remote

- Merged a custom ESLint rule into Oppia (6.8k stars) that fails CI on undocumented `ViewEncapsulation.None`, a recurring cause of CSS leaking across components; shipped 10 RuleTester tests and a 41-file legacy allowlist so CI stayed green.
- Earned merge access to review and land PRs for Oppia's Dev Workflow team after 2 PRs passed maintainer review.
- Moved FOSSology's Vagrant dev environment off end-of-life Ubuntu 20.04 to 22.04 to match its CI runners.

Research Intern | Birla Institute of Technology, Mesra

May 2026 – Jul 2026

NLP / Content Moderation | Python, scikit-learn, LightGBM, pandas | GitHub

- Built a 4-class comment-moderation classifier on 198k comments, combining word/char TF-IDF with 36 engineered signals (threat and identity lexicons, intent regexes, vote-controversy ratios) in an 80k-feature matrix.
- Trained Naive Bayes, L1 Logistic Regression and LightGBM with 5-fold stratified CV and class weights; the engineered features lifted a linear model from 0.54 to 0.82 macro-F1.
- Blended LightGBM and Logistic Regression with weights found by simulated annealing and per-class thresholds tuned with Nelder-Mead, reaching **0.836 macro-F1**, and raised F1 on the rare threat class (2.8% of data) from 0.23 to 0.65.

Projects

Loom | Multi-Agent LLM Orchestration Platform

Aug 2026 – Sep 2026

github.com/D3S-Gaurav/Loom | TypeScript, Next.js, Go, Kafka, gRPC, Redis, PostgreSQL, Temporal

- Built an LLM orchestrator whose planner emits a schema-validated task DAG; workers run independent tasks in parallel, and validator rejections trigger focused revisions (0–2 per run by mode).
- Capped LLM context cost with per-mode budgets on every hand-off (worker output, validator feedback, a 12k–48k-character synthesis corpus), shared fairly so one long answer cannot crowd out the rest.
- Treated attachments as untrusted input: GitHub fetches accept only exact repository URLs (no SSRF), and images reach agents only as vision-model text told to treat visible instructions as data.
- Streamed run events through Kafka and a Go service to a live graph UI over gRPC and WebSocket, with sequence-numbered replay so reconnecting clients never apply an event twice.

ChatFlow | Real-Time Group Chat

Jun 2026 – Jul 2026

github.com/D3S-Gaurav/chat-fullstack | TypeScript, Node.js, Express, Socket.IO, Prisma, PostgreSQL, k6

- Load-tested with k6 to **5,000 concurrent sockets** and **~97k message deliveries/s with zero loss** on one Node process (p95 40 ms at 1,000 sockets); traced the ceiling to the single-threaded broadcast.
- Built the backend on a typed Socket.IO event contract with Zod validation on every payload, JWT socket auth and role-based access; 32 integration tests run in CI against PostgreSQL 17.

Achievements

- Codeforces: rank 682 of 17,000+ (top 4%) in Round 1102 (Div. 2), max rating 1306; 600+ problems solved across LeetCode, Codeforces, CodeChef and GeeksforGeeks.
- Gold Level on WorldQuant BRAIN with 10,000+ points across 8 peer-reviewed alphas | Finalist, SEBI Hackathon.